

CATS⁺

Competishum Advance

Training Sessions

(Lecture - VI)

Ques Let P_k be a regular polygon of k sides each of whose sides has length unity.

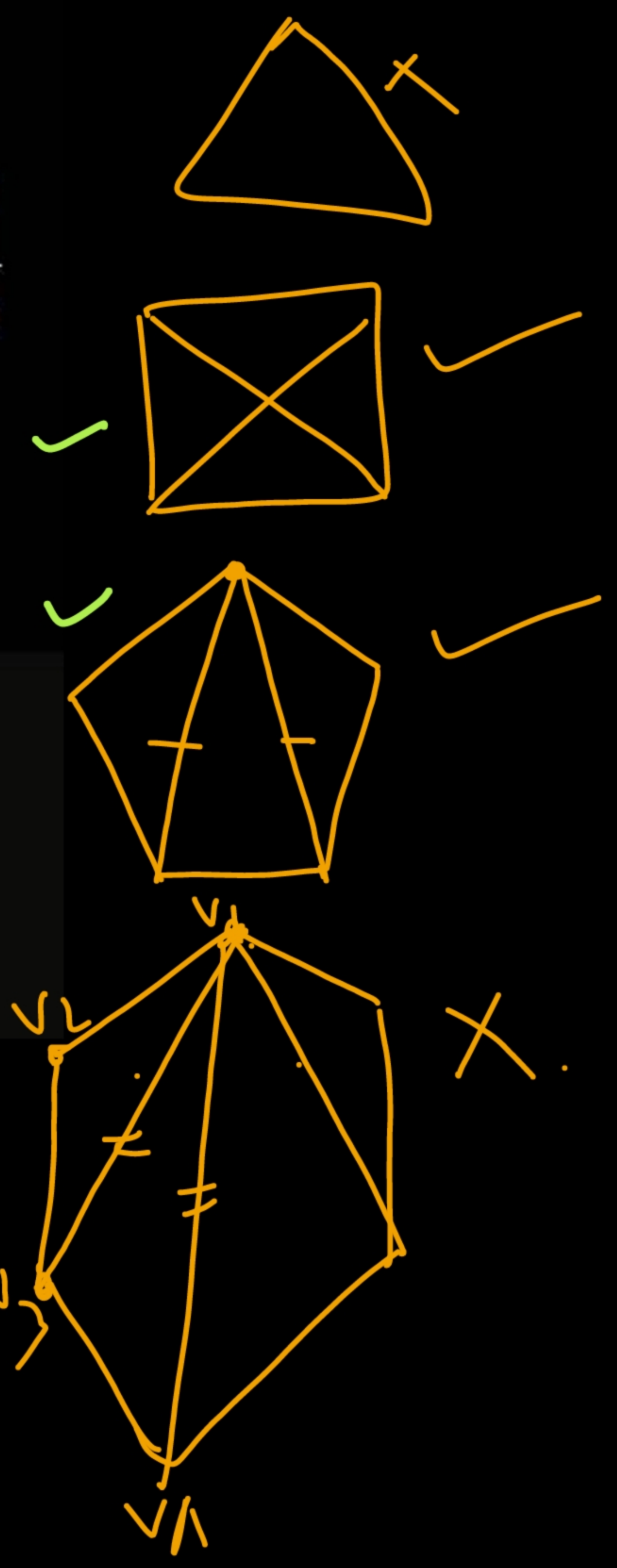
Consider set $A = \{P_3, P_4, P_5, \dots, P_{10}\}$ An element is chosen from set A ; then

(A) Probability that the all diagonals of the chosen polygon are equal in length is $\frac{1}{8}$

(B) Probability that all diagonals of the chosen polygon equal in length is $\frac{1}{4}$

(C) Probability that number of diagonals of chosen polygon is an odd number is $\frac{1}{2}$

(D) Probability that number of diagonals of chosen polygon is an odd number is $\frac{5}{8}$



$$P(\text{All diagonals are equal in length}) = \frac{2}{8} = \frac{1}{4} \Rightarrow \textcircled{B}$$

$$\left[\begin{aligned} & {}^nC_2 - n = \text{no. of diagonals} \\ &= \frac{n(n-1)}{2} - n \\ &= \frac{n(n-3)}{2} \end{aligned} \right]$$

$P = \frac{1}{2}$

$P_5, P_6, P_7, P_8, P_9, P_{10}$

Ques Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_7 \in \mathbb{R}$, and

$$v = \left(\sin(\alpha_1) + \sin(\alpha_2) + \dots + \sin(\alpha_7) \right)^2 + \left(\cos(\alpha_1) + \cos(\alpha_2) + \dots + \cos(\alpha_7) \right)^2$$

then v will be less than

☐ A 45

☒ B 50

☒ C 55

☒ D 60

Sol

$$V_n = \left(\sin(\alpha_1) + \sin(\alpha_2) + \dots + \sin(\alpha_n) \right)^2 + \left(\cos(\alpha_1) + \dots + \cos(\alpha_n) \right)^2$$

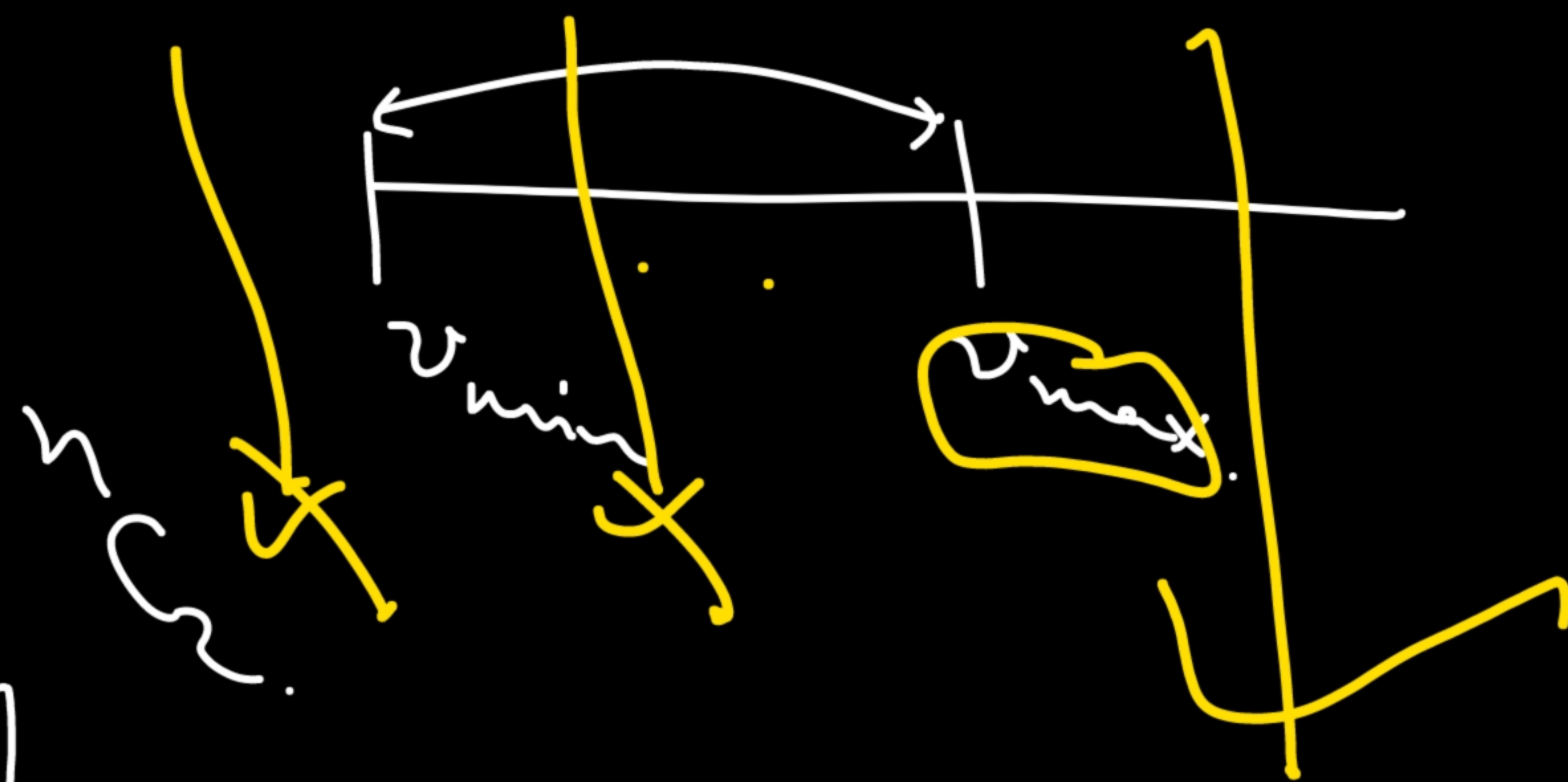
$$= \sum \sin^2(\alpha_i) + 2 \sum_{i>j} \sin(\alpha_i) \sin(\alpha_j) + \sum \cos^2(\alpha_i) + 2 \sum_{i>j} \cos(\alpha_i) \cos(\alpha_j)$$

$$= n + 2 \sum_{i>j} \cos(\alpha_i - \alpha_j)$$

$$\leq n + 2 \cdot {}^nC_2 = n + 2 \cdot \frac{n(n-1)}{2}$$

$$= n^2$$

$$V_7 \leq 49$$



A box contains 3 coins B_1 , B_2 and B_3 . Probability of getting head on these coins are $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$

Respectively. If one of the coins is selected at random and tossed it m times and got exactly 10 heads ($m \geq 10$, $m \in \mathbb{N}$) and x, y, z denotes the probabilities that it was coin B_1, B_2, B_3 respectively, then

(A) $x = \frac{4^{10}}{4^{10} + 2^{10} + 1}$ if $m = 10$

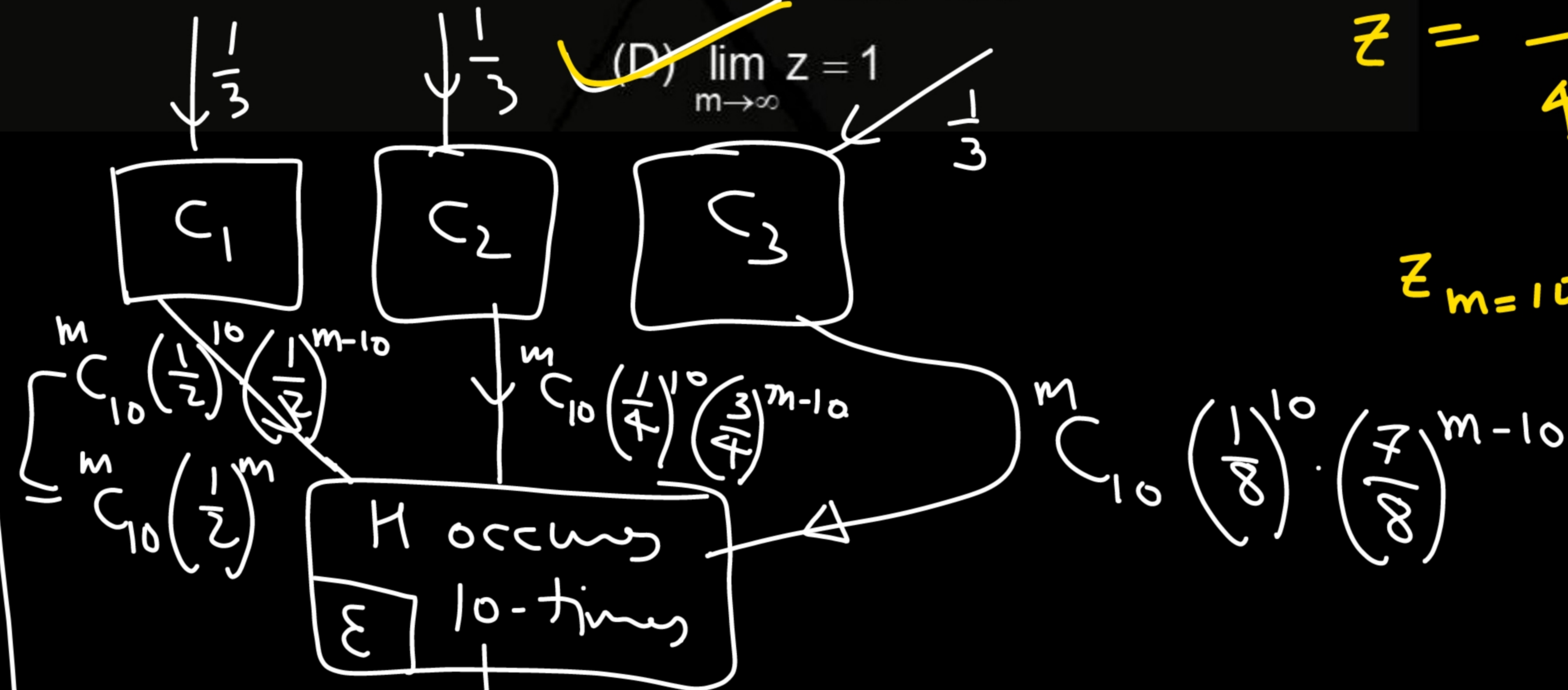
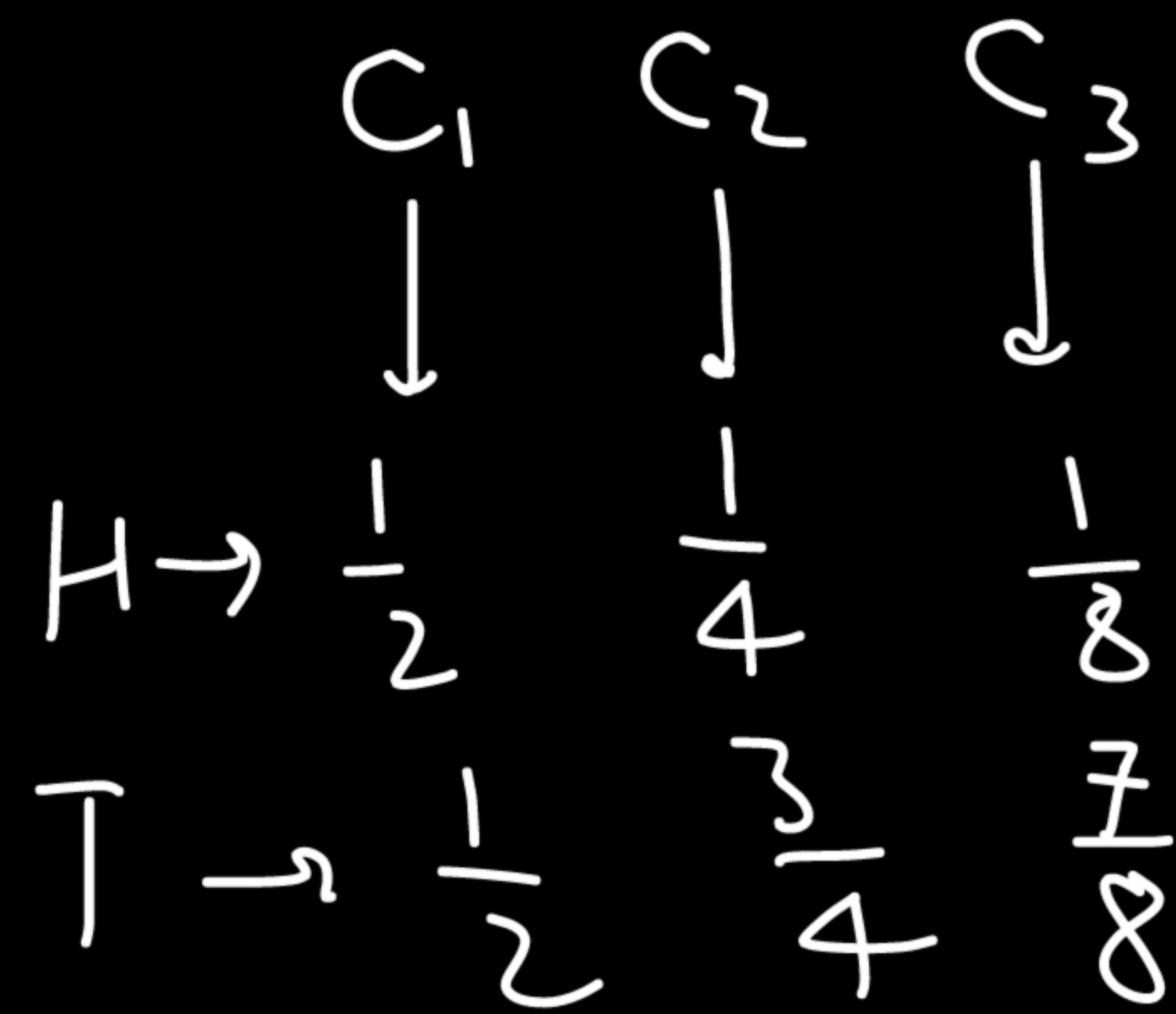
(B) $z = \frac{1}{4^{10} + 2^{10} + 1}$ if $m = 10$

(C) $\lim_{m \rightarrow \infty} x = 0$

(D) $\lim_{m \rightarrow \infty} z = 1$

$$z = \frac{7^{m-10}}{4^m + 3^{m-10} \cdot 2 + 7^{m-10}}$$

$$z_{m=10} = \frac{1}{4^{10} + 2^{10} + 1}$$



$$P(\xi) = \frac{1}{3} \left({}^m C_{10} \left(\frac{1}{2}\right)^m + {}^m C_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{m-10} + \left(\frac{1}{8}\right)^{10} \left(\frac{7}{8}\right)^{m-10} \right)$$

$$x = \frac{\frac{1}{3} \cdot C_{10} \cdot \left(\frac{1}{2}\right)^m}{\frac{1}{3} \cdot C_{10} \left(\left(\frac{1}{2}\right)^m + \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{m-10} + \left(\frac{1}{8}\right)^{10} \left(\frac{7}{8}\right)^{m-10} \right)}$$

$$= \frac{4^m}{8^m} + \frac{3^{m-10} \cdot 2^m}{8^m} + \frac{7^{m-10}}{8^m}$$

$$\frac{3^{m-10}}{4^m}$$

$$m=10$$

$$x = \frac{4^{10}}{4^{10} + 2^{10} + 1}$$

$$\frac{4^m}{4^m + \frac{1}{3^{10}} \cdot 6^m + \frac{1}{7^{10}} \cdot 7^m} = 0$$

$$= \frac{4^m}{4^m + 3^{m-10} \cdot 2^m + 7^{m-10}}$$

$$z = \frac{7^{m-10}}{4^m + 3^{m-10} \cdot 2^m + 7^{m-10}}$$

$$\lim_{m \rightarrow \infty} z = \frac{7^{10} \cdot \left(\frac{4}{7}\right)^m + \frac{7^{10}}{3^{10}} \cdot \left(\frac{6}{7}\right)^m}{1} = 1$$

Que If $A_1, A_2, A_3, \dots, A_{1006}$ be independent events such that $P(A_i) = \frac{1}{2^i}$ ($i = 1, 2, 3, \dots, 1006$) and probability

that none of the events occurs be $\frac{\alpha!}{2^\alpha (\beta!)^2}$, then:

$$\alpha = 2012$$

$$\beta = 1006$$

(A) β is of form $4k + 2$, $k \in \mathbb{I}$

(B) $\alpha = 2\beta$

(C) β is a composite number

(D) α is of form $4k$, $k \in \mathbb{I}$

$$P(A_i) = \frac{1}{2^i}, \quad i = 1, 2, 3, \dots, 1006.$$

$$P(A \cap \bar{B} \cap C) = P(A) \cdot P(\bar{B}) \cdot P(C)$$

$$P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \dots \bar{A}_{1006}) = P(\bar{A}_1) \cdot P(\bar{A}_2) \dots P(\bar{A}_{1006})$$

$$= \prod_{i=1}^{1006} \left(1 - \frac{1}{2^i}\right) = \prod_{i=1}^{1006} \left(\frac{2^i - 1}{2^i}\right)$$

$$= \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \dots \frac{2011}{2(1006)} = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \dots (2011)(2012)}{(2 \cdot 4 \cdot 6 \dots (2 \cdot 1006))^2}$$

$$= \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \dots 2011 \cdot 2012}{\left(2^{1006} \cdot (1 \cdot 2 \cdot 3 \dots 1006)\right)^2} = \frac{2012}{2^{2012} \cdot (1006)^2}$$

Ques Let length of the edge of the regular tetrahedron ABCD is '2'. Points E and F are taken on the edges AD and BD respectively such that E divides DA and F divides BD in the ratio 2:1 each. Then match the Column

Column - I

- (A) Length CF
- (B) Length EC
- (C) Area of $\triangle CEF$
- (D) Length EF

Column - II

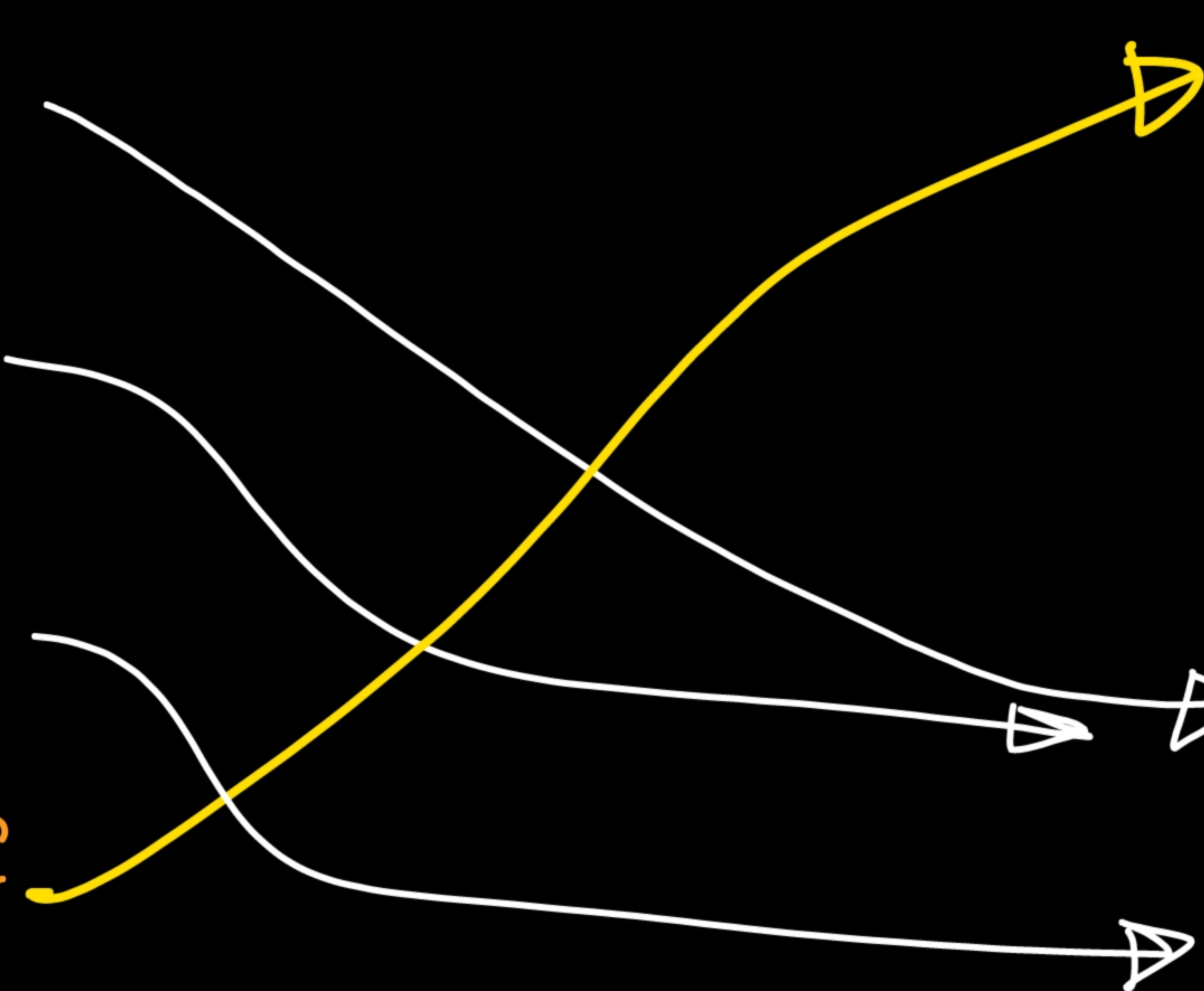
- (P) $\frac{2}{3}$
- (Q) $\frac{\sqrt{7}}{2\sqrt{3}}$
- (R) $\frac{5}{3\sqrt{3}}$
- (S) $\frac{2\sqrt{7}}{3}$
- (T) None of the above is correct Match

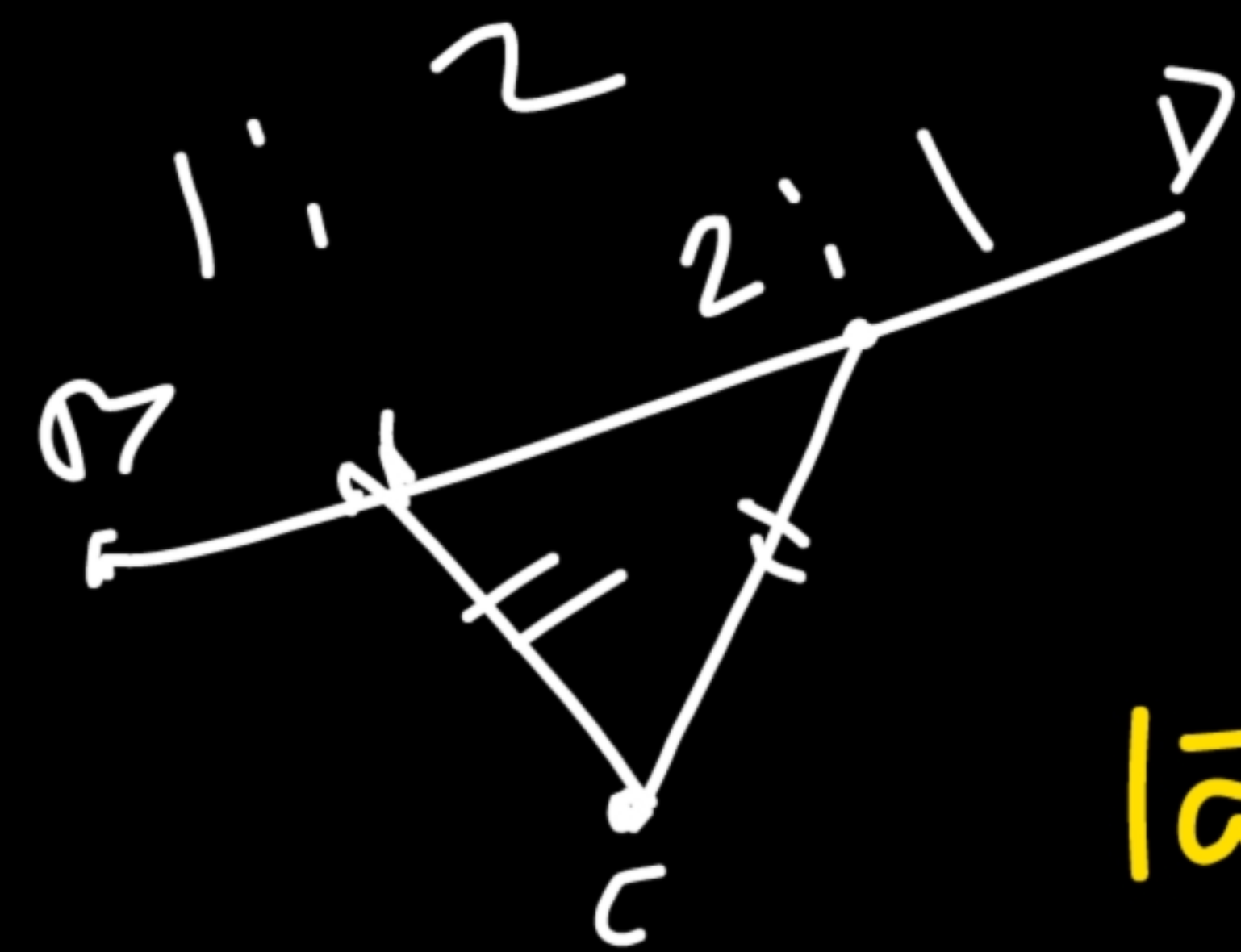
A $\rightarrow$ S

B $\rightarrow$ S.

C $\rightarrow$ T

D $\rightarrow$ P.





$$\textcircled{8} \quad (\hat{a}, \hat{b}) = (\hat{b}, \hat{c}) = (\hat{c}, \hat{a}) \\ = \frac{1}{3}$$

$$\overline{CF} = \overline{b} - \overline{c}$$

$$CF = \sqrt{\frac{b^2}{9} + c^2 - \frac{2b \cdot c}{3}}$$

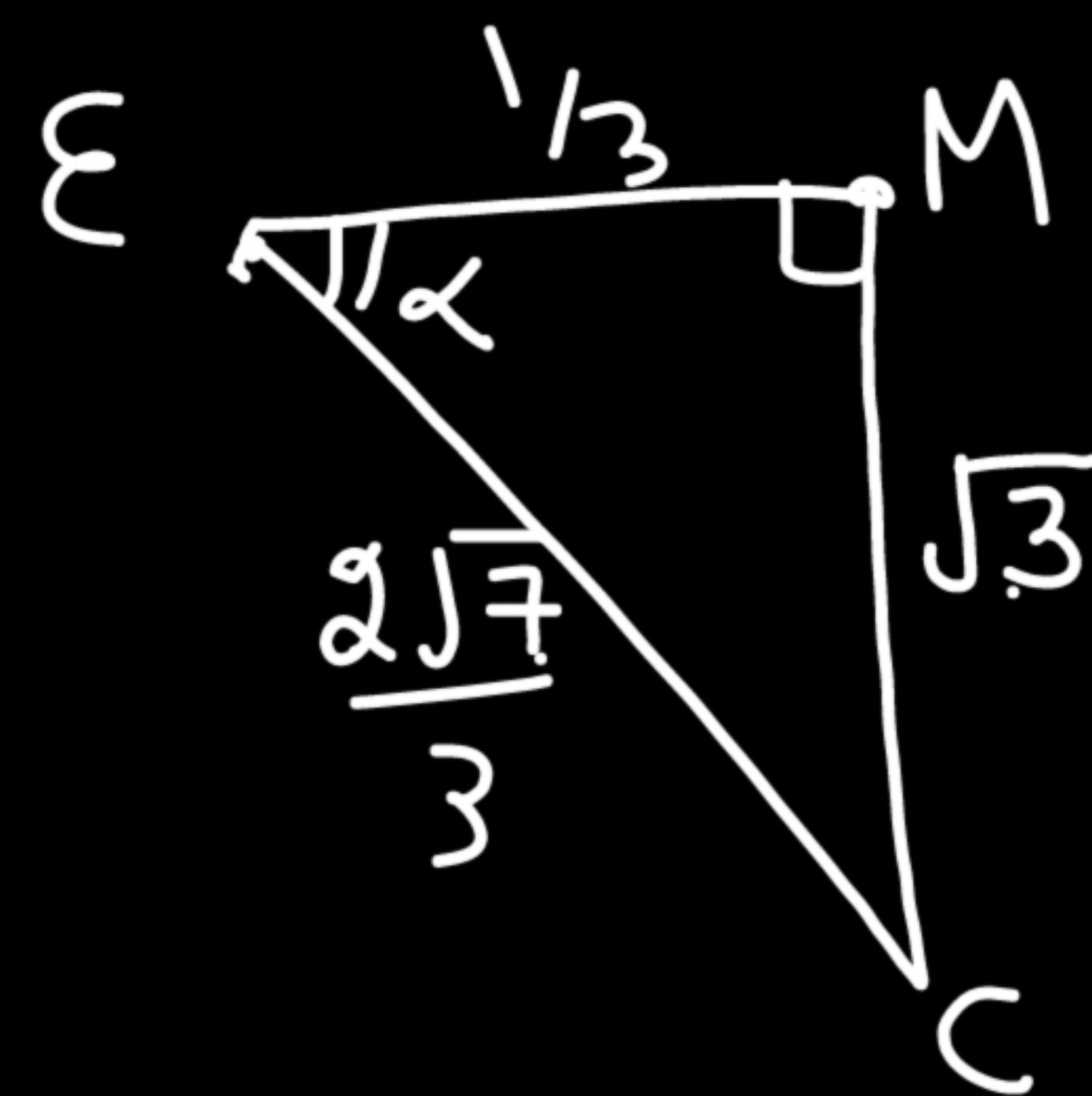
$$= \sqrt{\frac{4}{9} + 4 - \frac{2 \cdot 2}{3}}$$

$$\begin{aligned}\overrightarrow{CE} &= \frac{2}{3} \overline{a} - \overline{c} \Rightarrow |\overrightarrow{CE}| = \sqrt{\frac{4}{9} a^2 + c^2 - 2 \cdot \frac{2}{3} \overline{a} \cdot \overline{c}} \\ &= \sqrt{\frac{4}{9} \cdot 4 + 4 - \frac{8}{3}} \\ &= \sqrt{\frac{16 + 36 - 24}{9}} = \frac{2\sqrt{7}}{3}\end{aligned}$$

$$\begin{array}{r} CF \\ \hline \Sigma C \\ \Sigma F \\ \hline \Delta C \Sigma F \end{array}$$

$$\overrightarrow{CF} = \frac{1}{3}\overline{b} - \overline{c}$$

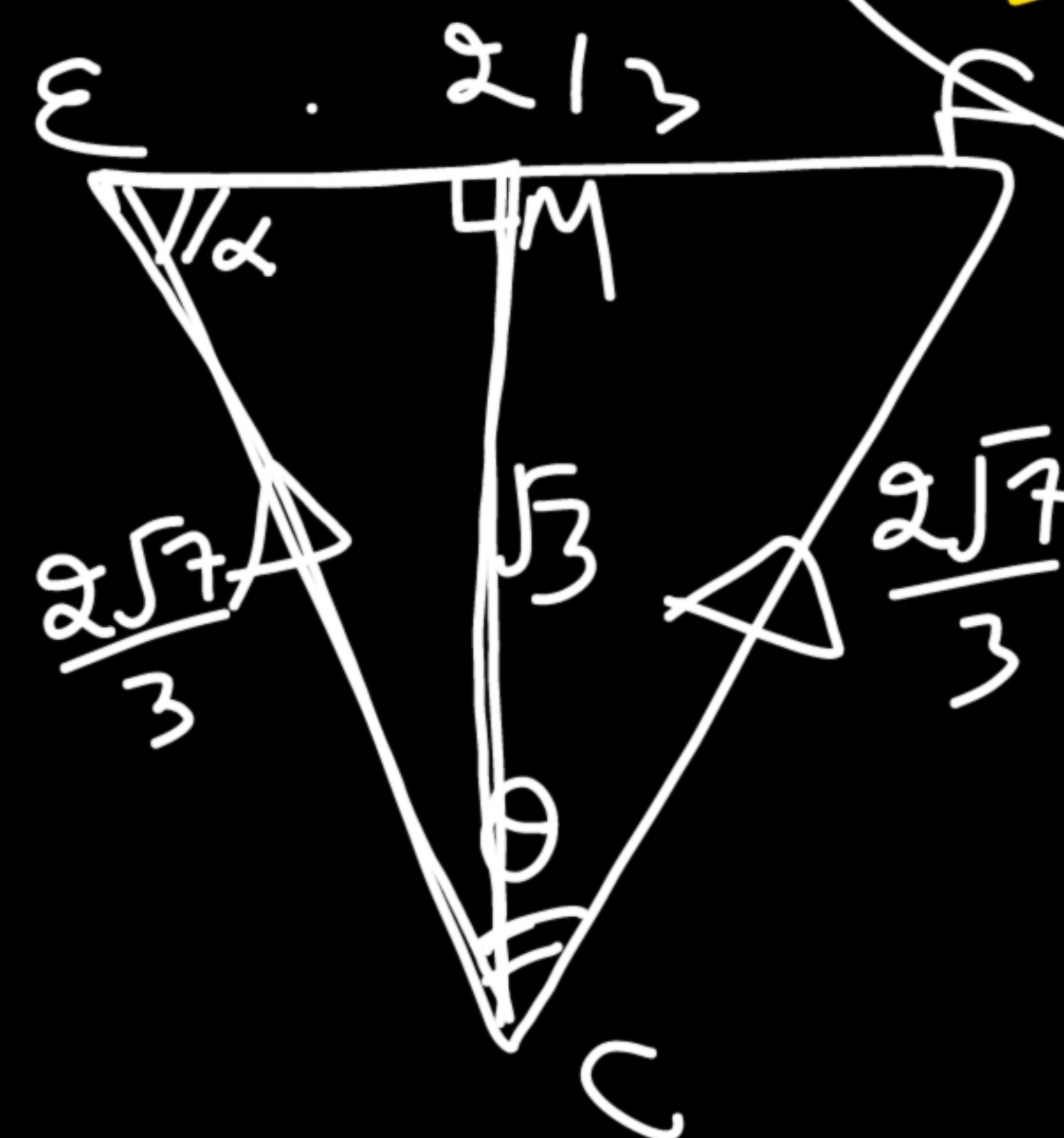
$$\overrightarrow{CE} = \frac{2}{3}\overline{a} - \overline{c}$$



$$\sqrt{\frac{28}{9} - \frac{1}{9}}$$

$$\frac{1}{2} \cdot \sqrt{3} \cdot \frac{2}{3} = \frac{1}{\sqrt{3}}$$

$$\sqrt{\frac{27}{9}} = \sqrt{3}$$



$$\frac{\overline{b} - 2\overline{a}}{3}$$

$$|\overrightarrow{EF}| = \sqrt{\frac{b^2 + 4a^2 - 4\overline{a} \cdot \overline{b}}{9}}$$

$$= \sqrt{\frac{1 + 16 - 8}{9}} = \frac{2}{3}$$

$$\text{Area} = \frac{1}{2} \cdot \sqrt{3} \cdot \frac{2}{3} = \frac{1}{\sqrt{3}}$$

Ques Let z is any complex number which satisfy $|z-2|=1$,
 also $\sin(\arg(z)) = \frac{(z-1)(z-\lambda)}{2|z|(2-z)} \cdot i$, where $\lambda \in \mathbb{N}$ then λ is
☒ A Odd integer ☒ B Prime Number ☐ C divisible by 5 ☐ D greater than 3.

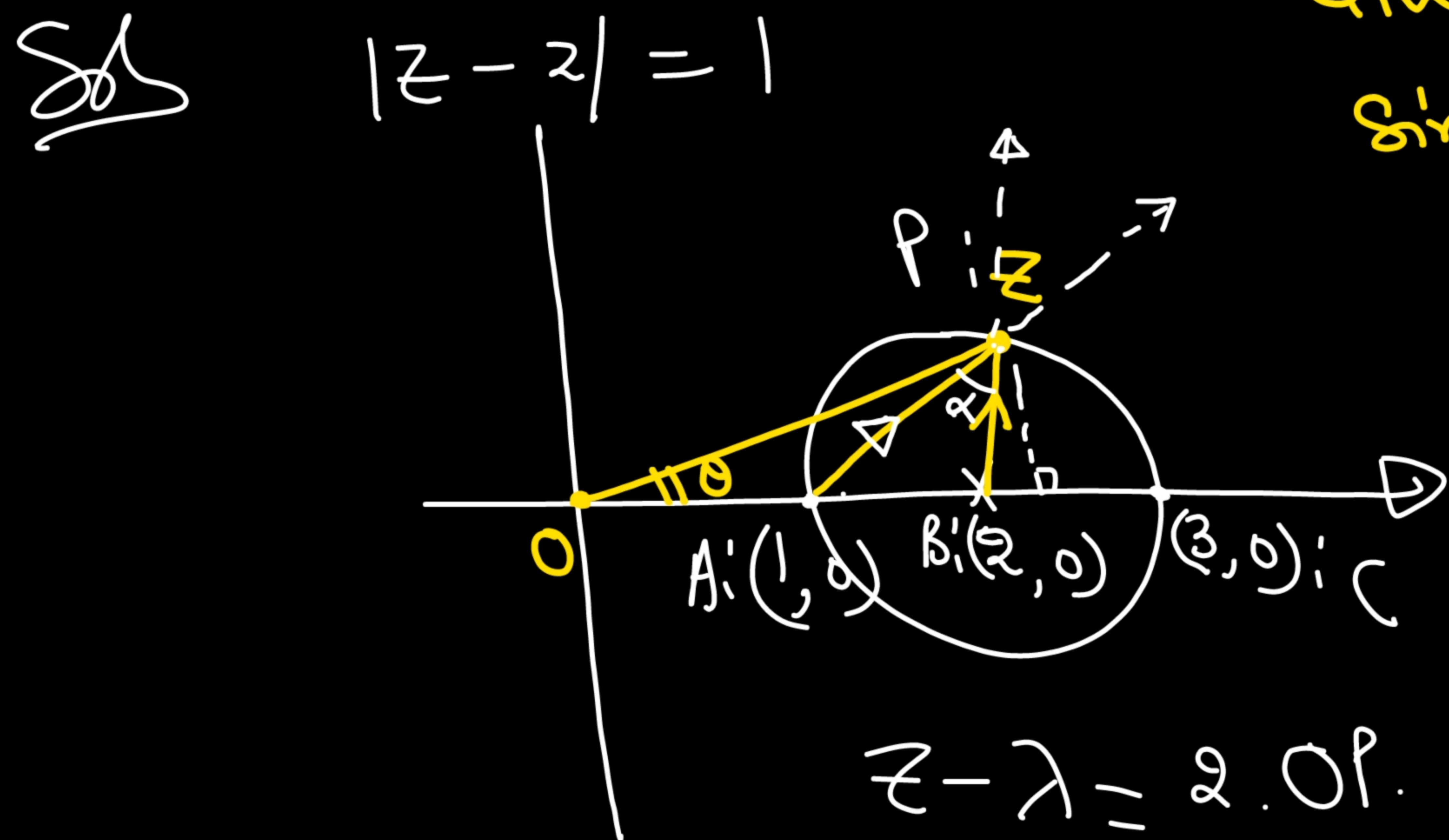
Given relation

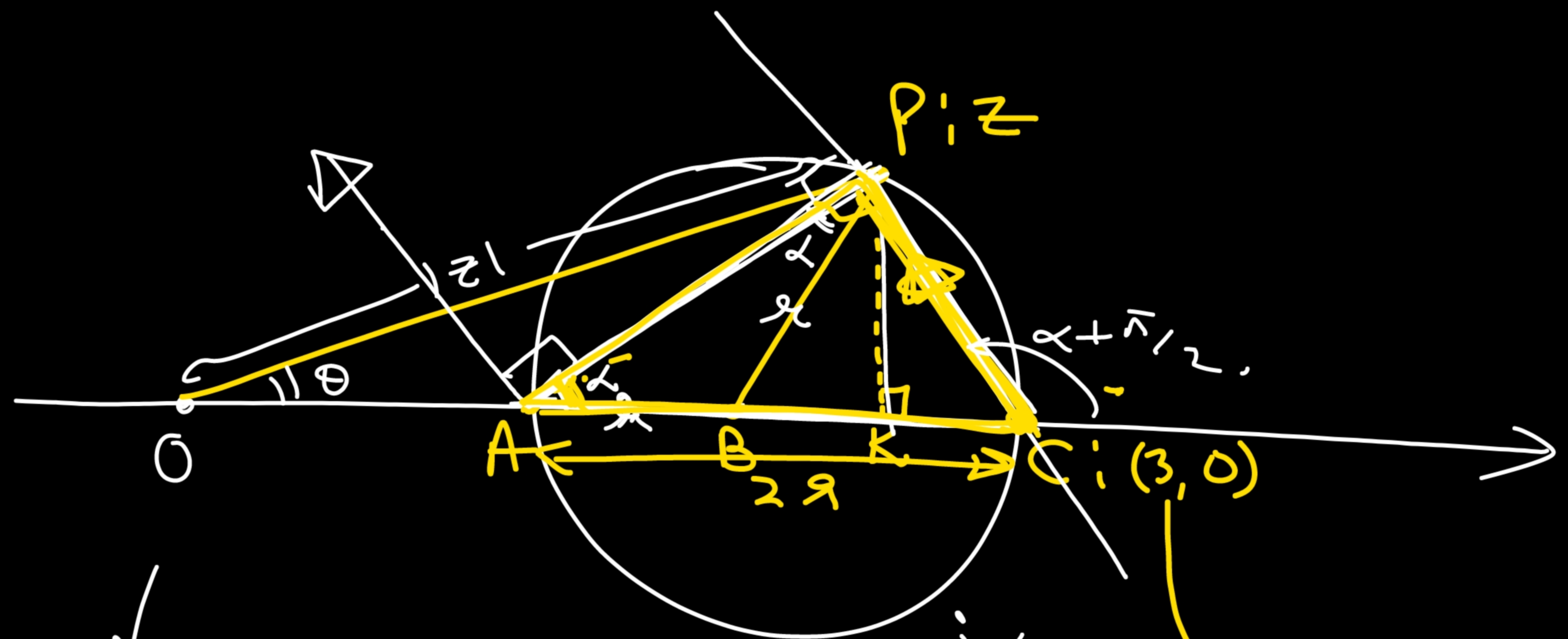
$$\sin(\theta) = \frac{(z-1)(z-\lambda)}{2|z|(2-z)} \cdot i \Rightarrow (z-\lambda) = \frac{2|z| \cdot \sin(\theta)}{(z-2)} \cdot i$$

$$z-\lambda = 2 \cdot |z| \cdot \left(\frac{z-2}{z-1} \right) \cdot \sin(\theta) \cdot i$$

$$\frac{PB}{PA} \cdot e^{i\alpha}$$

$$z-\lambda = 2 \cdot OP \cdot \frac{PB}{PA} \cdot \sin(\theta) \cdot i \cdot e^{i\alpha}$$





$$\begin{aligned}
 \underline{z - \lambda} &= 2|z| \cdot \sin(\theta) \cdot \frac{PB}{PA} \cdot i \cdot e^{i\alpha} \\
 &= 2 \cdot \underbrace{PK}_{\substack{AB \\ PA}} \cdot e^{i(\alpha + \frac{\pi}{2})} \\
 &= \underbrace{2h \cdot \sin(\alpha)}_{\perp PC} \cdot e^{i(\alpha + \frac{\pi}{2})} = \boxed{PC} \cdot e^{i(\alpha + \frac{\pi}{2})}
 \end{aligned}$$

$\lambda = 3$

Ques Let z_1, z_2, z_3 be the complex numbers such that $|z_1-1|=|z_2-1|=|z_3-1|$ and $\arg\left(\frac{z_3-z_1}{z_2-z_1}\right)=\frac{\pi}{6}$, then the value of $|\underbrace{z_2(z_2-1)-z_3(z_2+1)+(z_3+1)(z_3-1)}|$

- is
- ☒ A Even integer ☒ B Prime Number

Δ $|z_1-1|=|z_2-1|=|z_3-1|$

$\hookrightarrow (1,0)$ is circumcentre
of Δ with vertices
 z_1, z_2, z_3

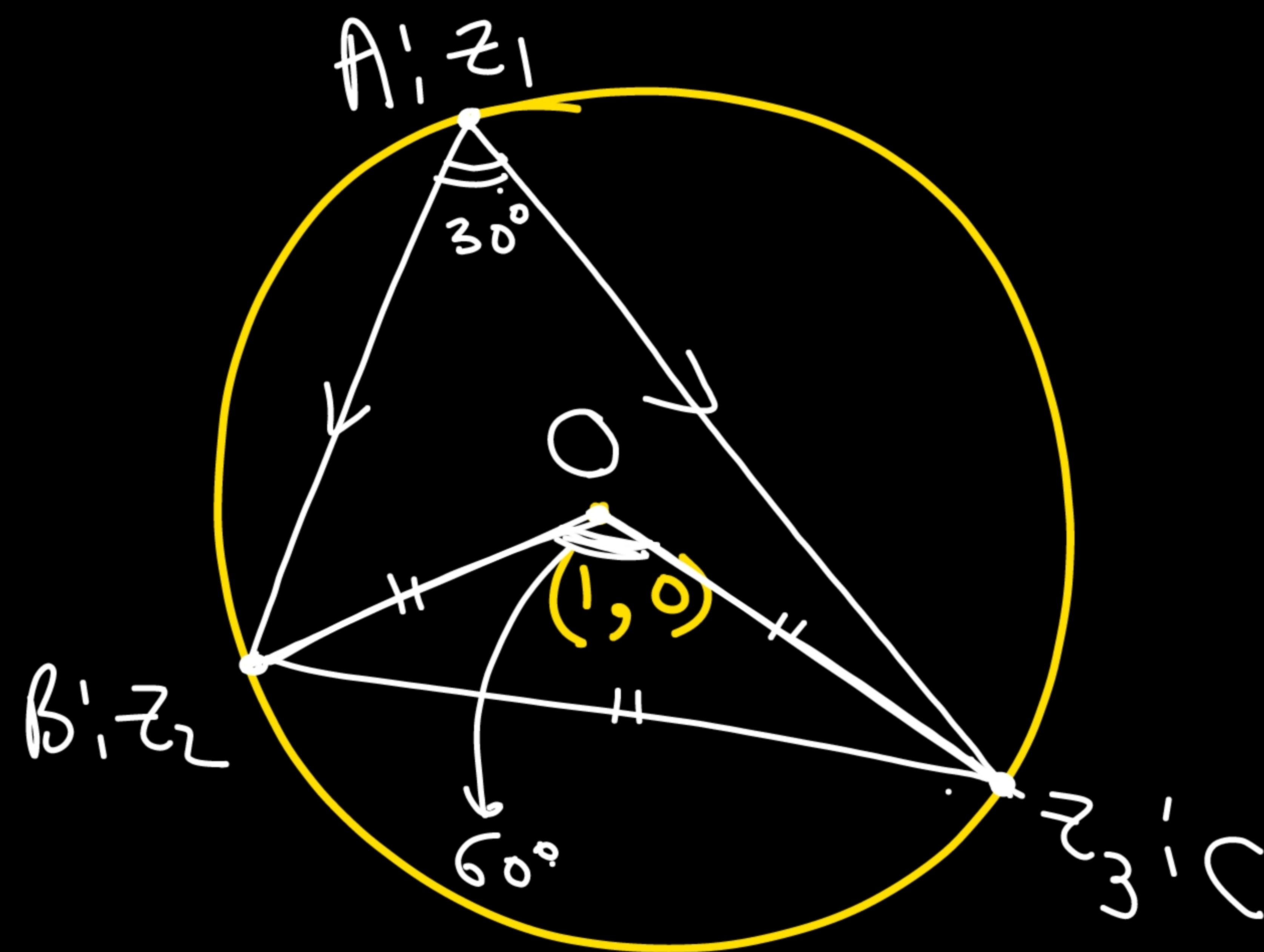
☐ C > 3 ☒ D ≤ 5

$$= |z_2^2 - z_2 - z_2 z_3 - z_3 + z_3^2 - 1|$$

$$= |z_2^2 + z_3^2 - z_2 z_3 - z_2 - z_3 - 1|$$

$$= |\underbrace{z_2^2 + z_3^2 - z_2 z_3 - z_2 - z_3 + 1}_0 - 2|$$

$$= \boxed{2}$$



Clearly $\triangle OBC$ is an equilateral $\triangle$

$$a^2 + b^2 + c^2 = ab + bc + ca.$$

$$a = 1, b = z_2, c = z_3$$

$$1^2 + z_2^2 + z_3^2 = 1 \cdot z_2 + z_2 z_3 + 1 \cdot z_3$$

$$\underline{z_2^2 + z_3^2 - z_2 - z_3 - z_2 z_3 + 1 = 0}$$

Paragraph Let S is $(x-2)^2 + (y-3)^2 = 25$, with centre C and P is a point with coordinates $(5, -1)$, Now another circle S_1 is drawn with PC as diameter. Now Answer the followings! -

Q-1 If PQ is a variable chord of S such that it is bisected by S_1 then, how many such points Q exists?

- ☐ A 0 ☐ B 2 ☒ C Infinite ☐ D None of these.

Q-2 If PQ is a variable chord of S such that $\triangle CPQ$ is an equilateral $\triangle$ then, how many such points Q exists?

- ☐ A 0 ☒ B 2 ☐ C Infinite ☐ D None of these.

$$S \rightarrow (x-2)^2 + (y-3)^2 = 25$$

$P(5, -1) \rightarrow$ lies on the circle.

